

# Task Tunnel

## UX Findings and Milestone Planning Report

A consolidated handoff of product-design decisions, competitor research, interaction rules, information architecture, and milestone dependencies for the Android MVP.

### Core product thesis

Task Tunnel protects the intention that caused someone to open a distracting app while preserving the useful parts of that app. It is not primarily an app blocker.

### SOURCE OF TRUTH

Task Tunnel Solo-Dev Feasibility Dossier, current to 20 September 2026. Where this report adds UX behavior not explicitly specified in the dossier, it is labeled as a Design Recommendation. Competitor research is used to inform interaction decisions, not to override the dossier.

**DOSSIER REQUIREMENT** Required by the source dossier

**DESIGN RECOMMENDATION** Recommended UX behavior based on product reasoning/research

**POST-MVP IDEA** Useful later, not required for v0.1

**UNNECESSARY SCOPE** Explicitly avoid for the current MVP

# Executive summary

The UX should make the user feel that their original intention has more staying power - not that another app is controlling their phone.

- The central problem is intention continuity: users often open Instagram or YouTube for a legitimate task and are pulled into nearby recommendation surfaces they did not intend to use.
- The main runtime experience happens over the native apps: Purpose Gate -> Task Tunnel -> Surface Intervention -> Drift Detection -> Attention Debugger.
- Successful on-task use should be visually quiet. Task Tunnel should largely disappear while the user stays within the declared purpose.
- Surface interventions must preserve three legitimate choices: Return, End Tunnel, and Allow Anyway.
- Intentional Browse is a first-class choice. Leisure is not treated as failure.
- Detector uncertainty must fail open; the product should prefer missing an intervention over falsely blocking legitimate use.
- The main app should center two destinations: Attention and Protection. Settings remain secondary.
- The Attention Debugger should explain behavioral sequences rather than become another screen-time dashboard.
- Current beta hypotheses: temporary surface allowance after Allow Anyway (~5 min) and quick-return Tunnel grace (~60 sec). These are test values, not dossier requirements.



## 1. UX problem and product mental model

### DOSSIER REQUIREMENT

**Task Tunnel is solving an intention-continuity problem, not a raw screen-time problem.** The user should be able to keep Instagram and YouTube available for useful tasks while receiving friction when they move into surfaces that conflict with the purpose they just declared.

#### Correct mental model

"I opened Instagram to reply. Task Tunnel remembers that. If I reflexively enter Reels, it reminds me that Reels is outside the task - then lets me choose what happens next."

- Do not frame all app use as bad.
- Do not frame every override as failure.
- Do not attempt to make bypass impossible.
- Do not infer that a user has "finished" a task unless they explicitly end the Tunnel or a defined session rule applies.
- Do not expose AccessibilityService mechanics to ordinary users.

## 2. Product layers

### DESIGN RECOMMENDATION

| Layer   | Purpose                                   | Examples                                                                  |
|---------|-------------------------------------------|---------------------------------------------------------------------------|
| Setup   | Establish trust and configuration         | Onboarding, Accessibility disclosure, privacy, supported apps, Drift pool |
| Runtime | Protect intention while another app is in | Purpose Gate, active Tunnel, surface intervention,                        |

|            |                                                  |                                                                        |
|------------|--------------------------------------------------|------------------------------------------------------------------------|
|            | use                                              | expiry, Drift check-in                                                 |
| Reflection | Help the user understand what happened afterward | Attention Debugger, episode history, rolling trends, protection health |

**Design implication:** Task Tunnel should not optimize for time spent inside its own app. Its core value happens at the boundaries of Instagram and YouTube.

### 3. Onboarding and trust journey

#### DOSSIER REQUIREMENT



**The Accessibility permission should not be the first thing the user sees.** The user first needs to understand the product benefit and why the permission is necessary.

| User question             | Required answer                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------|
| What can Task Tunnel see? | Enough of the visible app interface to classify supported surfaces such as Reels or Shorts. |
| Why does it need this?    | To tell whether the user has moved outside the Task Tunnel they created.                    |
| What is stored?           | Only task/session/surface events required for the product and Attention Debugger.           |
| What is not stored?       | No screenshots and no persisted private message content in the normal path.                 |
| Where is data processed?  | Local-first on the device for the MVP; no account or cloud backend required.                |

#### DESIGN RECOMMENDATION

**Drift Detection setup should be skippable during onboarding.** Purpose Gate + Task Tunnel are the central concept; Drift is an additional mechanism and should not make first-run setup feel heavy.

### 4. Purpose Gate

#### DOSSIER REQUIREMENT

The Purpose Gate asks what the user opened a supported app to do. Choices must be task-shaped, fast, and morally neutral.

| Instagram MVP        | YouTube MVP                       |
|----------------------|-----------------------------------|
| Reply to messages    | Search / watch something specific |
| Browse intentionally | Browse intentionally              |

#### POST-MVP IDEA

Additional examples from the broader dossier - Post/share, Find a specific account/post, Subscriptions, Upload/manage - should not silently become v0.1 requirements.

#### DESIGN RECOMMENDATION

##### Gate interaction rule

Ask one primary question: "What did you open this app for?" Task selection is primary; optional duration is secondary. Avoid turning the gate into a miniature settings form.

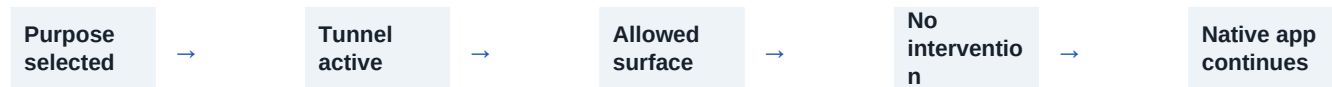
- Keep the number of choices small enough to scan immediately.

- Browse intentionally must have equal visual dignity to task-oriented options.
- Session duration is optional; do not force a timer onto every Tunnel.
- A repeated Purpose Gate on every short app resume would quickly become annoying and should be prevented by Tunnel grace behavior.

## 5. Active Task Tunnel

### DOSSIER REQUIREMENT

When the user stays on an allowed surface, Task Tunnel should not intervene.



### DESIGN RECOMMENDATION

**Avoid persistent focus UI.** No floating timer, glowing border, or motivational bubble is needed while the user is on-task. A brief start confirmation is enough, after which Task Tunnel should disappear.

## 6. Surface intervention and action semantics

### DOSSIER REQUIREMENT

#### Canonical intervention

“You opened Instagram to reply. Reels is outside this Task Tunnel.”

Actions: Return • End Tunnel • Allow Anyway

**The tone should produce “Oh right” rather than “You are not allowed.”** Avoid alarm-red unless there is a real error.

| Action       | User meaning                            | Recommended behavior                                                            |
|--------------|-----------------------------------------|---------------------------------------------------------------------------------|
| Return       | My original intention still stands      | Navigate back toward an allowed surface without exposing Android mechanics.     |
| End Tunnel   | My original intention no longer applies | End the Tunnel, but do not automatically close the host app.                    |
| Allow Anyway | I knowingly want this detour            | Temporarily allow the incompatible surface; preserve the larger Tunnel context. |

## 7. Allow Anyway duration

### DOSSIER REQUIREMENT

Every intervention must have a legitimate user-controlled override. Overrides should not be treated as failure.

### COMPETITOR-INFORMED DESIGN RECOMMENDATION

**Use a temporary surface allowance.** Comparable products use temporary unlock/pause windows rather than forcing an immediate repeated prompt or turning one override into a permanent disable.

- one sec uses time-based access and re-intervention patterns.
- ScreenZen uses short usage bursts and repeated reflection.
- AppBlock allows restrictions to be paused for selected durations when strict blocking is not enabled.
- Jomo supports temporary unlocks and optional strictness rather than making strictness universal.

#### Initial beta hypothesis

Start testing with approximately 5 minutes of temporary allowance after “Allow Anyway.” This value is a design hypothesis, not an experimentally proven optimum.

#### UNNECESSARY SCOPE

Do not ask “5 / 10 / 20 minutes?” every time the user taps Allow Anyway. That makes a one-tap autonomy choice unnecessarily bureaucratic.

## 8. Tunnel lifetime and quick-return grace

#### COMPETITOR-INFORMED DESIGN RECOMMENDATION

A Tunnel should survive a short, legitimate switch to another app but should not remain stale indefinitely.

| Event                                                | Result                                                      |
|------------------------------------------------------|-------------------------------------------------------------|
| User taps End Tunnel                                 | End immediately                                             |
| Timed session expires and user chooses to finish     | End immediately                                             |
| User leaves protected app and returns quickly        | Resume the same Tunnel                                      |
| User leaves protected app beyond grace               | End; next open gets a new Purpose Gate                      |
| Timed Tunnel expires while host app is in background | End silently; do not throw an overlay over an unrelated app |

#### Initial beta hypothesis

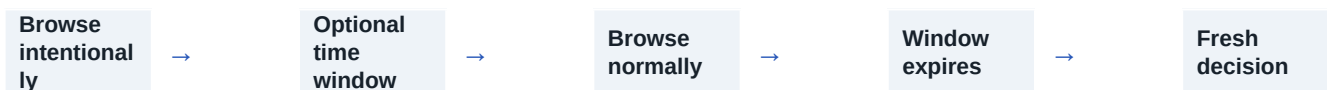
Start around a 60-second return-grace window. one sec documents a similar intentional app-switching mechanism and uses one minute as an example interval. Treat this as a testable starting value, not a fixed truth.

## 9. Intentional Browse and session expiry

#### DOSSIER REQUIREMENT

Intentional leisure is a legitimate user choice. “Browse intentionally” must not be hidden, visually downgraded, or described as giving in.

#### DESIGN RECOMMENDATION



**Expiry should re-intervene rather than force-close the app.** The interaction should simply return the decision to the user.

- Conceptual choices: Finish, Keep browsing, Choose another purpose.
- Avoid “limit exceeded,” guilt language, warning red, or punitive copy.
- If the user chose Browse intentionally with no duration, do not secretly impose one.

## 10. Drift Detection

### DOSSIER REQUIREMENT

Drift Detection groups rapid movement among selected distraction apps into a possible single attention episode.

#### Example pattern

Instagram → Reddit → YouTube within roughly one minute, potentially indicating that the user is hunting for stimulation after abandoning the original task.

### DESIGN RECOMMENDATION

**Drift should be softer than a surface mismatch.** A Reels mismatch is deterministic relative to a Reply Tunnel; cross-app switching is only a behavioral signal and may be legitimate.

| Principle                 | MVP behavior                                                    |
|---------------------------|-----------------------------------------------------------------|
| Intervention strength     | Soft check-in, not a hard block                                 |
| Tone                      | Observation / question, not accusation                          |
| Possible actions          | Set an intention; Keep going                                    |
| Frequency                 | One check-in per detected episode                               |
| Automatic Tunnel creation | No                                                              |
| Initial beta trigger      | About 3 selected apps within ~60 seconds, subject to validation |

#### Tone example

“Looking for something? You’ve moved between Instagram, Reddit and YouTube in under a minute.”

## 11. Attention Debugger

### DOSSIER REQUIREMENT

The Attention Debugger should explain how a distraction episode unfolded, not merely how long the phone was used.

| Time | Event                       |
|------|-----------------------------|
| 7:22 | Opened Instagram to message |
| 7:27 | Entered Reels               |
| 7:50 | Switched to Reddit          |
| 8:04 | Opened YouTube              |
| 8:31 | Episode ended               |

### DESIGN RECOMMENDATION

Build the debugger around four semantic event types:

| Event type   | Meaning                                   |
|--------------|-------------------------------------------|
| Intent       | What the user explicitly said they wanted |
| Transition   | How app/surface activity changed          |
| Intervention | Where Task Tunnel surfaced awareness      |
| Decision     | What the user chose in response           |

The visual identity should come from making causal sequence and choices legible - not from decorating timestamps or adding generic charts.

## 12. Progress and metrics

### DOSSIER REQUIREMENT

Avoid streak obsession and purity metrics. A bad day should not reset the user to zero.

| Metric                    | Recommended treatment                                                        |
|---------------------------|------------------------------------------------------------------------------|
| Intentional-session rate  | Suitable for a quiet rolling summary                                         |
| Drift episodes per week   | Suitable for a quiet rolling summary                                         |
| Trend direction           | Useful when paired with context                                              |
| Overrides                 | Usually more informative inside individual episodes than as a headline total |
| Abandoned automatic opens | Potentially useful summary metric; validate before emphasizing               |
| Raw screen time           | Secondary at most; not the identity of the product                           |

### UNNECESSARY SCOPE

Do not add a “focus score,” daily grade, or giant productivity KPI unless later evidence shows a clear user benefit.

## 13. Fail-open behavior and reliability

### DOSSIER REQUIREMENT

**Detector uncertainty must fail open.** If the system cannot confidently identify a surface, it should allow the user through rather than risk blocking a legitimate task.

| Internal reality                               | Normal-user translation                                |
|------------------------------------------------|--------------------------------------------------------|
| Service running; detectors healthy             | Protection active - everything is working normally     |
| Persistent detector degradation / app redesign | Limited protection - some interventions may not appear |
| Accessibility permission disabled              | Protection off - Accessibility access needs attention  |
| Single UNKNOWN surface                         | No runtime warning; continue normally                  |

### UNNECESSARY SCOPE

- Do not show detector confidence scores.
- Do not show resource IDs or accessibility node classes.
- Do not show raw fingerprints or node counts to normal users.

## 14. Information architecture

### DESIGN RECOMMENDATION

**Avoid the conventional blocker/productivity navigation of Home / Focus / Insights / Profile.** There is no account, and users should not need to open Task Tunnel to start a Tunnel.

| Primary destination  | Question it answers                                   | Core contents                                                                       |
|----------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------|
| Attention            | What happened to my attention?                        | Today, episodes, Debugger timelines, trends, history                                |
| Protection           | Is Task Tunnel protecting me the way I configured it? | Status, Instagram, YouTube, Drift Detection, compatibility/health                   |
| Settings (secondary) | How should Task Tunnel itself behave?                 | Session defaults, intervention preferences, privacy/data, notifications, help/about |

### Default destination

Attention should be the default main-app destination because the Attention Debugger is the product's most

distinctive reflective surface. Protection status can still appear as a quiet line at the top when needed.

## 15. Attention screen and episode detail

### DESIGN RECOMMENDATION

**The Attention screen should be chronological first and analytical second.** Do not make it a card-filled dashboard.

- Show recent Task Tunnels and drift episodes as the hero content.
- Show a small number of rolling trends below the timeline rather than a wall of KPI cards.
- Episode detail gets its own screen and should show intention, transitions, interventions, and decisions in sequence.
- An active Tunnel may appear contextually at the top if the user manually opens Task Tunnel while it is running; it does not need a permanent navigation destination.

## 16. Protection and per-app configuration

### DESIGN RECOMMENDATION

Protection should answer three questions in order: Is it working? What is protected? How is it configured?

| Area              | Recommended model                                               |
|-------------------|-----------------------------------------------------------------|
| Overall status    | Protection active / Limited protection / Protection off         |
| Instagram         | Task choices such as Reply to messages and Browse intentionally |
| YouTube           | Task choices such as Search/watch and Browse intentionally      |
| Drift Detection   | On/off, included apps, brief explanation                        |
| Protection health | Human-language service/app compatibility state                  |

**Per-app configuration should be intention-centric.** Do not make “Block Reels: ON / Block Explore: ON” the primary mental model. The user configures intentions; the product owns the corresponding policy.

## 17. Visual and motion direction

### DOSSIER / USER DIRECTION

| Aim for                                | Avoid                                        |
|----------------------------------------|----------------------------------------------|
| Modern Android                         | Generic AI-product aesthetic                 |
| Dark theme with restrained blue accent | Purple/blue gradient overload                |
| Clean typography and strong hierarchy  | Glassmorphism everywhere                     |
| Minimal visual clutter                 | Card soup and meaningless dashboards         |
| Subtle depth and premium restraint     | Glowing focus widgets                        |
| Calm, slightly technical, human        | Cyberpunk, preachy, juvenile, corporate SaaS |

### DESIGN RECOMMENDATION

**Motion should communicate state, not decorate the interface.** Useful moments include Tunnel start, intervention appearance, return to intended task, and Attention Debugger transitions. Avoid dramatic warning animation.

## 18. Production UI vs developer UI

### DOSSIER REQUIREMENT

| Production language                 | Developer-only language             |
|-------------------------------------|-------------------------------------|
| Protection active                   | detector confidence                 |
| YouTube / Instagram                 | AccessibilityNodeInfo               |
| You're in a Task Tunnel             | resource IDs / node classes         |
| Shorts is outside your current task | structural signals / detector state |
| 1 drift episode today               | raw fingerprints / node count       |

Developer diagnostics may exist in hidden/dev builds, but technical complexity should remain invisible to ordinary users.

## 19. Competitor research takeaways

| Product   | Relevant lesson for Task Tunnel                                                                                                                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| one sec   | Time-based access, re-intervention, feature blocking, and intentional app-switching grace show that temporary access and repeated reflection are mature patterns.          |
| ScreenZen | Pause-before-open and short usage bursts validate friction at boundaries; friction alone is not differentiation.                                                           |
| AppBlock  | Feature-level blocking and temporary schedule pauses are already established; strict anti-bypass should not be Task Tunnel's primary value proposition.                    |
| Jomo      | Temporary unlocks and optional strict modes reinforce that different users want different strictness; strictness should remain optional rather than philosophical core.    |
| UNDOOMED  | Closest philosophical analogue: preserve useful parts while filtering feeds. Task Tunnel differentiates by temporary task-dependent native policies plus Drift + Debugger. |

### Competitive conclusion

Borrow mature patterns for timing, grace, temporary escape, and re-decision - but keep Task Tunnel differentiated by task-dependent surface policy and the end-to-end loop of intention → protection → drift → explanation.

## 20. Current behavioral decisions for planning

| UX question                | Current MVP direction                                         |
|----------------------------|---------------------------------------------------------------|
| Allow Anyway               | Temporary surface allowance                                   |
| Initial allowance duration | Test around 5 minutes                                         |
| Tunnel ending              | Explicit End, expiry decision, or leaving beyond return grace |
| Quick app switch           | Preserve current Tunnel                                       |
| Initial return grace       | Test around 60 seconds                                        |
| Quick reopen               | Reuse Tunnel inside grace; re-gate after grace                |
| Intentional Browse         | First-class, non-shaming option                               |
| Browse expiry              | Neutral re-decision; never force-close                        |
| Drift intervention         | Soft check-in, once per episode                               |
| Active Tunnel visibility   | Brief confirmation, then mostly invisible                     |
| Detector uncertainty       | Fail open                                                     |
| Main app default           | Attention                                                     |
| Primary IA                 | Attention + Protection; Settings secondary                    |

Important: the ~5 minute allowance and ~60 second return grace are beta hypotheses, not dossier requirements. They should remain configurable/testable in planning.

## 21. MVP / post-MVP / unnecessary scope

### DOSSIER REQUIREMENT - MVP

- Android only
- Instagram + YouTube
- Accessibility onboarding and protection status
- Purpose Gate
- Small task set
- Task Tunnel policy
- Reels/Explore and Shorts detection
- Return / End Tunnel / Allow Anyway
- Optional session window and expiry re-intervention
- Drift Detection
- Local Attention Debugger
- A few rolling metrics
- Local-first privacy; no account/cloud backend

### POST-MVP IDEA

- Additional task types
- Additional supported apps
- Browser escape detection
- Advanced intervention personalization
- Longer history / advanced insights
- Custom intervention library
- Downloaded detector rule packs
- Additional locales

### UNNECESSARY SCOPE FOR MVP

- ML/LLM coach
- Generic all-app support
- Cloud accounts/sync
- Hard anti-uninstall system
- Complex gamification/streaks/achievements
- Automatic task-completion inference
- Persistent focus widgets
- Social/community features
- Large analytics dashboard

## 22. Milestone design dependencies

| Milestone               | Design work that should be settled                                                                                                                                                      |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M3 - Task Tunnel        | Purpose Gate; Tunnel state model; Return / Allow Anyway / End semantics; quick-switch grace; Browse intentionally; session expiry; active-Tunnel status; intervention hierarchy/copy    |
| M4 - Drift              | Drift check-in UX; episode grouping model; included-app selection; one-check-in-per-episode behavior                                                                                    |
| M5 - Attention Debugger | Attention main screen; event taxonomy; timeline; episode detail; rolling metrics; empty states                                                                                          |
| M6 - Hardening / Play   | Protection health; permission repair; unsupported app-version states; OEM/background failure states; privacy/disclosure polish; production/dev UI separation; Play-review support flows |

## 23. Principles the milestone planner should preserve

1. Protect intention, not abstinence.
2. Intentional leisure is legitimate.
3. Friction should interrupt automaticity, not punish.
4. Every intervention has a legitimate escape hatch.
5. Overrides must not look like failure.
6. Successful use should be visually quiet.
7. Do not expose detector machinery to normal users.
8. Fail open when uncertain.
9. Show reliability clearly without technical jargon.
10. Explain attention behavior rather than merely measuring time.
11. Do not build the product around streaks or purity.
12. Keep concepts and components small enough for one developer to implement and maintain consistently.

## 24. Planner-ready condensed brief

### Handoff summary

Task Tunnel should behave like a quiet runtime layer over Instagram and YouTube. The user declares a purpose through a fast Purpose Gate, uses the native app normally, and sees Task Tunnel only when they enter a surface incompatible with that declared purpose. The intervention offers Return, Allow Anyway, and End Tunnel. Allow Anyway is currently planned as a temporary surface allowance; short app switches preserve the current Tunnel for a grace window. Intentional Browse is legitimate and session expiry returns the decision to the user rather than force-closing the app. Drift Detection is a softer, once-per-episode check-in. The main app centers Attention (timeline/Debugger) and Protection (configuration/reliability), with Settings secondary. The visual language should be calm, minimal, premium, and Android-native. The product must fail open when detector certainty is insufficient.

## References

Primary product source

- Task Tunnel Solo-Dev Feasibility Dossier, 15 pages, research and feasibility review current to 20 September 2026.

Competitor / pattern research used in the UX decisions

one sec - Immediate / Delayed / Always Ask intervention modes: <https://tutorials.one-sec.app/en/articles/3974530>

one sec - Intentional App Switching: <https://tutorials.one-sec.app/en/articles/3310146>

one sec - Intervention troubleshooting / grace behavior: <https://tutorials.one-sec.app/en/articles/3262274>

ScreenZen - Google Play feature description: <https://play.google.com/store/apps/details?id=com.screenzen>

AppBlock - Android schedules / pause behavior: <https://appblock.app/help/android/schedules/>

Jomo - Help / strictness model: <https://help.jomo.so/>

UNDOOMED - feature-level filtering / Messages Only model: <https://undoomed.app/>